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# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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**SDC HACKATHON 2026**

# Technical & Tools Declaration

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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | High Five |
| **Project / Solution Name** | CareerLens AI |
| **Name (Leader)** | Lai Jie Ying |
| **Email & Phone (Leader)** | +60 11-2097 0647, jieying.lai47@gmail.com |
| **University Affiliation(s)** | Universiti Tunku Abdul Rahman (UTAR) |
| **Live Prototype URL** | |
| **Code Repository URL (public / access granted to judges)** | |

## 2. Architecture & Technology Stack

### 2.1 System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

1. **User Interface:** Job seekers and HR professionals interact with their respective dashboards to upload CVs, complete skills assessments, explore job opportunities, and access workforce insights. User inputs are transmitted to the Flask backend through RESTful APIs.

2. **Backend:** Flask processes incoming requests, extracts CV information, manages authentication, and coordinates skills matching, course recommendations, and workforce analysis. It retrieves relevant data and communicates with AI services when required.

3. **AI Models and APIs:** Gemini API and fine-tuned Qwen2.5 models generate career guidance, interview assessments, and strategic workforce recommendations. Similarity-based algorithms match skills with jobs and courses, while predictive models estimate occupational disruption and future talent demand. Results are returned through Flask to the dashboards.

4. **Database:** SQLite stores user profiles, applications, and access permissions. Structured datasets, including O*NET, DOSM, and JobStreet, provide occupational, salary, and employment information to support backend processing and AI recommendations.

5. **Cloud Infrastructure:** Replit supports web application deployment and real-time testing, while RunPod provides GPU resources and JupyterLab for AI model fine-tuning. GitHub supports version control and collaboration.

### 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| Python 3.11 | Core backend runtime, data processing pipelines, and AI orchestration. | PSF License |
| Flask (v3.0.x) | Lightweight web backend framework powering 50+ RESTful API endpoints, session security, and role-based access control. | BSD-3-Clause |
| Vanilla JavaScript (ES2022) & HTML5/CSS3 | High-performance, zero-bloat client-side rendering for the dual-portal interactive workspaces, future radar, and interview buddy. | W3C / ECMA Standard |
| Tailwind CSS (v3.x) | Utility-first CSS framework for fluid responsive layouts, typography, and HuaShu glassmorphism styling. | MIT License |
| Leaflet.js (v1.9.4) | Interactive geospatial mapping engine rendering 68k+ job nodes and candidate proximity radius filters. | BSD-2-Clause |
| Chart.js (v4.x) | Client-side interactive data visualizations for 2030 workforce trends and HR skill heatmaps. | MIT License |
| pypdf (v4.x) & fpdf2 (v2.7.x) | PDF text extraction, anonymized CV structure parsing, and automated applicant profile PDF report generation. | BSD-3-Clause / LGPLv3 |
| Waitress (v3.0.2) | Production-ready WSGI HTTP server powering multi-threaded request serving in cloud deployment. | ZPL-2.1 |
| SQLite 3 | Lightweight transactional relational database for user accounts, applications, and audit logs. | Public Domain |

| Name & Version | Purpose in Project | License |
| --- | --- | --- |
| openpyxl (v3.1.x) & pandas (v2.x) | Extraction, hygiene, and structuring of official DOSM 2025 Excel tables into optimized JSON caches. | MIT / BSD-3-Clause |

## 3. APIs & Cloud Services

### 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| Google Gemini API (gemini-3.5-flash / gemini-2.5-flash) (Google Cloud / Google AI Studio) | Powers Lumi AI Career Advisor, dynamic 3-question competency verification, and interactive resume co-writing. | Free Tier / Sponsor-provided |
| Remotive Remote Jobs API (Remotive.com) | Real-time public streaming data feed of global cloud and remote tech job postings for the Future Skills Radar. | Free / Public Open API |
| CARTO Basemaps & OpenStreetMap Tile API (CARTO / OpenStreetMap) | Renders street-level geographic basemap tiles within Leaflet for job proximity discovery. | Free / Open Access |
| Nominatim Geocoding API (OpenStreetMap) | Resolves user coordinates and address lookups into Malaysian cities and administrative districts. | Free / Open Access |

### 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| **Service & Provider** | **Purpose in Project** | **Plan / Region** |
|---|---|---|
| Replit (PaaS Cloud Runtime) | Cloud hosting and continuous 24/7 web application deployment running Flask + Waitress on standard web ports. | Free / Core Tier (Global Edge / US-Central) |
| RunPod (GPU Cloud Compute) | On-demand cloud GPU pods (NVIDIA RTX 4090 / A100) used for fine-tuning our Qwen LoRA model weights via Unsloth. | Community Cloud / On-demand (US / EU) |
| GitHub | Version control repository hosting, multi-branch synchronization and automated test runners. | Free Tier (Global) |

## 4. AI Tools, Models & Datasets

### 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| Gemini 3.5 Flash & 2.5 Flash | Google | Pre-trained (called via API with domain grounding) | Google Generative AI Terms of Service |
| careerlens_lumi_lora (Base: Qwen2.5-7B-Instruct) | Alibaba Cloud / Custom fine-tuned | Fine-tuned (PEFT/LoRA adapter trained for career advisory & anti-hallucination) | Apache 2.0 (Base) / MIT (Adapter) |
| interview-adapter (Base: Qwen2.5-1.5B-Instruct) | Alibaba Cloud / Custom fine-tuned | Fine-tuned (trained on NVIDIA RTX 4000 Ada for interview question branching & assessment) | Apache 2.0 (Base) / MIT (Adapter) |
| careerlens_totem_lora (Base: Qwen2.5-7B-Instruct) | Alibaba Cloud / Custom fine-tuned | Fine-tuned (trained on RunPod via Unsloth on O*NET 30.3 & Big Five personality data) | Apache 2.0 (Base) / MIT (Adapter) |
| Course Recommender & Skill Gap Ranker (v1.0) | Self-developed | Self-trained (TF-IDF vectorizer + Cosine Similarity ranking over 5,014 verified courses) | MIT License |

### 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| DOSM Salaries & Wages Survey Report 2025 | Department of Statistics Malaysia (DOSM) (https://open.dosm.gov.my/publications/salaries_wages_2025) | Public (Official Government) | Open Data Government of Malaysia | N |
| JobStreet Malaysia All Job Dataset | Kaggle (azraimohamad/jobstreet-all-job-dataset) (68,447 jobs, 20,514 companies) | Public | CC0: Public Domain | N |
| O*NET 29.0 & 30.3 Database & Occupation Taxonomy | U.S. Department of Labor / O*NET Resource Center (https://www.onetcenter.org/) | Public | CC BY 4.0 | N |
| AI Impact on Job Market (2024–2030) Dataset | Kaggle (30,000 occupational risk projections) | Public / Synthetic Research | CC BY-SA 4.0 | N |
| AI Workforce Displacement Global (2020–2026) | Kaggle / Academic Research (20,800 sector shift records) | Public | Open Database License (ODbL) | N |
| EdX University Courses Dataset 2021 | Kaggle (720 verified university courses with skills metadata) | Public | CC0: Public Domain | N |
| IBM HR Analytics Employee Attrition & Performance | IBM Research / Kaggle (1,470 employee records) | Public / Synthetic Benchmark | Apache 2.0 | N |
| Gender Pay Gap & Wage Equity Benchmarks | U.S. Census Bureau CPS/PSID (25,000 statistical rows) | Public | Public Domain | N |
| Updated Resume Dataset | Kaggle (gauravduttakiit/resume-dataset) (962 labeled CVs across 25 domains) | Public | CC BY-SA 4.0 | N |

| | | | | |
|---|---|---|---|---|
| Synthetic Demo Candidate Resumes | Internal Project Synthetic Data (src/static/sample_cvs/) | Synthetic (fictional personas e.g., Aisha Rahman) | MIT License | N |
| Company Reviews, Salaries, Location Dataset | Kaggle — https://www.kaggle.com/datasets/hardikchoudharyhc/company-reviews-salaries-location-dataset | Public | CC BY-NC-SA 4.0 | N |
| Glassdoor Job Reviews | Kaggle — https://www.kaggle.com/datasets/davidgauthier/glassdoor-job-reviews | Public | CC BY-SA 4.0 | N |
| Glassdoor Job Reviews 2 | Kaggle — https://www.kaggle.com/datasets/davidgauthier/glassdoor-job-reviews-2 | Public | CC BY-NC-SA 4.0 | N |
| Company Reviews and Ratings from AmbitionBox | Kaggle — https://www.kaggle.com/datasets/nagrajdesai/company-reviews-and-ratings-from-ambitionbox | Public | CC0: Public Domain | N |
| Software Engineering Interview Questions Dataset | Kaggle — https://www.kaggle.com/datasets/syedmharis/software-engineering-interview-questions-dataset | Public | MIT | N |

### 4.3 AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

| **Tool & Version** | **Used For** | **Extent of Use (minor / moderate / major)** |
|---|---|---|
| Google Antigravity (Gemini-powered Agentic Coding Assistant) | RESTful API architecture, dataset extraction scripts, multi-branch Git synchronization, and automated test suite creation | Major |
| OpenAI Codex | Helped develop frontend interfaces and backend functions, adjust layouts for different devices, connect APIs, identify coding errors, and improve existing code. | Major |
| Midjourney v6 / DALL·E 3 | Helped create UI illustrations, Totem Spirit mascot characters, and Interview Buddy graphics | Moderate |
| GitHub Copilot / Code Completion Tools | Suggested code while programming and assisted with location-mapping functions, skill extraction patterns, and CSS styling. | Minor |
| ChatGPT (GPT-4o) | Helped create fictional candidate profiles for demonstration purposes and review content related to SDG 4 and SDG 8.<br>Assisted in writing and debugging Python scripts for dataset cleaning, processing, and JSON file generation | Minor |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

Our team reviewed and refined AI-generated outputs before integrating them into CareerLens AI. For AI-assisted coding, team members inspected generated code, corrected errors, and tested frontend interfaces, backend functions, and API integrations.

For data processing, we verified dataset sources, checked Python scripts, and validated the accuracy and structure of processed data files. AI-generated documentation and SDG-related content were reviewed against project requirements and supporting references.

We also conducted functional testing, debugging, and cross-device testing to identify issues. All AI-generated outputs underwent human review, and our team took full responsibility for the final implementation and submission.

## 5. Sponsor-Provided Resources

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

| **Sponsor** | **Resource (API / credits / platform / data)** | **How It Was Used in the Project** |
|---|---|---|
| Replit | Cloud Development Platform | Used to develop and improve responsive frontend interfaces, preview changes in real time across different devices, and integrate Git for code synchronisation and team collaboration. |
| RunPod | GPU Cloud Computing Platform / JupyterLab | Used to access a GPU-enabled JupyterLab environment for training and fine-tuning Qwen2.5 AI models, experimenting with training workflows, and troubleshooting model development. |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

Yes, the sponsor-provided resources played an important role in developing CareerLens AI. Replit enabled our team to develop and test responsive interfaces efficiently through real-time previews and Git integration. RunPod provided GPU computing resources and JupyterLab for training and fine-tuning AI models, which would otherwise require additional computing infrastructure. These resources significantly reduced technical barriers, accelerated our development process, and enabled us to apply the skills gained during SDC Phase 1 workshops. Although alternative platforms exist, Replit and RunPod were valuable in supporting the successful development of our prototype.

## 6. Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

CareerLens AI collects names, email addresses, password hashes, profile information, uploaded CVs, job applications, feedback, chat prompts, and access logs. Data is stored in SQLite and application-managed files, with passwords securely hashed and CVs kept private by default.

Employers can only access CV summaries or documents explicitly shared by users. Access permissions can be revoked, and sharing grants expire. With user consent, extracted CV information may be sent to RunPod for AI processing, while CV summaries and interview responses may be shared with Google Gemini for interview assessments. Chat prompts may also be sent to configured AI providers.

Users can delete uploaded CVs, extracted information, related applications, and access history. However, automatic data retention schedules and account deletion are not yet implemented. Encryption at rest remains unverified, and previously downloaded CV copies cannot be remotely deleted.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

CareerLens AI has several technical limitations and risks. Generative AI models such as Gemini and Qwen2.5 may produce inaccurate career advice, interview evaluations, or workforce recommendations. To reduce hallucinations, the system grounds AI responses using structured datasets such as O*NET and DOSM.

Candidate screening and job matching may introduce bias due to incomplete datasets or differences in CV formatting. Skills-based matching and competency assessments help reduce reliance on CV keywords, although bias cannot be eliminated entirely.

The AI Job Disruption Radar and workforce forecasts are estimates based on historical data and simulations rather than guaranteed predictions. External AI APIs may also experience rate limits, latency, or service interruptions. Local processing and precomputed datasets reduce dependence on external services.

Privacy risks are addressed through user consent, controlled CV sharing, and revocable access permissions.

## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[  ]

[24/10/2026]